

Taleef Tamsal

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Research interests: LLM safety and adversarial robustness; reliable evaluation; agentic AI security; trustworthy NLP/RAG.

Education

Purdue University Fort Wayne

Aug. 2024 – May 2026

M.S. in Computer Science, GPA: 4.00/4.00

Fort Wayne, IN

Advisor: Dr. Jonathan Rusert **Thesis:** *Do Efficient Adaptations Reduce Safety? Jailbreak Robustness of PEFT vs. Full Fine-Tuning on Consumer-Accessible LLMs.* [\[thesis\]](#)

Lahore University of Management Sciences (LUMS)

Aug. 2020 – May 2024

B.S. in Computer Science

Lahore, Pakistan

Publications

All entries are first-authored.

Peer-Reviewed Archival Papers

1. “Adaptation, Not Algorithm: LoRA and Full Fine-Tuning Show Comparable Black-Box Jailbreak Degradation in Five Open-Weight LLMs.”
Taleef Tamsal and Jonathan Rusert.
Findings of the Association for Computational Linguistics: ACL-IJCNLP 2026. Accepted. [\[code/artifacts\]](#)
2. “PFW Task 8 at SemEval-2026 Task 8: Lightweight Tri-Fusion Retrieval with Prompt-Engineered Faithful Generation for Multi-Turn RAG.”
Taleef Tamsal and Jonathan Rusert.
Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval 2026), pp. 1526–1532, ACL, 2026.
[\[paper\]](#) [\[code\]](#)
3. “PFW at SemEval-2026 Task 6: Multi-Seed DeBERTa Ensembles for Political Response Clarity and Evasion Classification.”
Taleef Tamsal and Jonathan Rusert.
Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval 2026), pp. 1518–1525, ACL, 2026.
[\[paper\]](#) [\[code\]](#)

Peer-Reviewed Workshop Paper (Non-Archival)

4. “Your Judge Is a Confound: Evaluator and Attack Choice Distort Jailbreak-Safety Measurement for Fine-Tuned LLMs.”
Taleef Tamsal and Jonathan Rusert.
First Workshop on Reliable Evaluation for Language Models (JUDGE), NeurIPS 2026. Accepted for oral presentation; also accepted at AdvML-Frontiers × CoTMA, COLM 2026. [\[artifacts\]](#)

Research Experience

Graduate Researcher, LLM Safety and Evaluation

Sep. 2025 – May 2026

Purdue University Fort Wayne, advised by Dr. Jonathan Rusert

Fort Wayne, IN

- Ran a controlled study of how LoRA, QLoRA, 4-bit quantization, and full fine-tuning affect jailbreak robustness across five open-weight LLM families. The evaluation covered six attacks on HarmBench and JailbreakBench, with three automated judges checked against 750 blind human annotations. [\[code/artifacts\]](#)
- Used a pre-specified ± 5 pp TOST to show that LoRA and full fine-tuning had equivalent black-box safety degradation under the controlled recipe. QLoRA was more variable, while 4-bit quantization alone showed no meaningful degradation.
- Found that evaluator choice could change the conclusion as much as the adaptation method. GPT-4o-mini mislabeled 45.7% of human-consensus safe outputs as jailbreaks, compared with 8.6% for HB-CLs, and inflated measured degradation by about $3\times$.
- For the JUDGE study, tested whether the evaluator problem persisted with stronger judges and repeat scoring. GPT-5.6 configurations still over-flagged 21.5–29.9% of safe completions, and identical outputs received different labels on 3–15% of repeat evaluations.

Research Collaborator, Query-Efficient Adversarial NLP

June 2025 – Aug. 2025

Purdue University Fort Wayne, Dr. Jonathan Rusert’s group

Fort Wayne, IN

- Studied where black-box text attacks spend their query budget by separating selection from replacement in methods such as TextFooler and TextBugger. Compared datasets, victim models, and attack traces to identify where fewer queries could achieve the same attack objective.

Ongoing Research

Validity Instruments in Adversarial Attack Evaluation	2026 – Present
- Building a budget-aware black-box attack agent against hate-speech classifiers, and testing whether the standard validity constraints used to score such attacks decide which attacker strategy appears strongest. - Finding that constraints inherited from word-substitution attacks measure edit size rather than whether the harmful content survived, so they reject rewriting attackers by construction; calibrating alternative survival-based judges against human-written contrast pairs. - Seeded and query-cached experiments with bootstrap intervals, a frozen design and deviation log, and human review of sampled outputs.	

Selected Shared-Task Research

UNLP 2026, Ukrainian Multi-Domain Document Question Answering	2026
Built an offline Ukrainian document-QA pipeline with BGE-M3 retrieval, Qwen3 reranking, and quantized MamayLM-Gemma-3-12B generation to fit the shared task’s 9-hour compute budget.	
UNLP 2025, Social-Media Manipulation Detection	2025
Built an explainable classifier for 10 manipulation techniques and span-level detection over 9.5K Ukrainian Telegram posts, with interpretable rationales for a heavily imbalanced label set.	
SemEval-2025 Task 10, Multilingual Narratives [code]	2024 – 2025
Built XLM-R/BERT systems for entity-role framing, narrative classification, and grounded narrative extraction across five languages, reaching about 8× the shared baseline on internal validation.	

Industry Experience

Software Developer	May 2026 – Present
<i>Medical Informatics Engineering</i> [WorkWell] - Building WorkWell Measure Studio as its sole developer, a TypeScript/Next.js platform that runs CMS FHIR/CQL measure logic, reads FHIR R4 data through SMART Backend Services, and keeps the evidence behind each computed result. - Made measure correctness a release gate. Nine CMS measures now pass all 455 official MADiE test cases, the backend has 3,268 automated tests, and a read-only MCP interface exposes 13 role-gated tools without allowing AI to determine compliance.	
Development Intern, NLP and Retrieval-Augmented Generation	May 2025 – Aug. 2025
<i>Medical Informatics Engineering</i> [RASS] - Built and containerized the first RASS prototype for semantic search over 3,000+ Redmine issues, combining document ingestion, embeddings, OpenSearch retrieval, and LLM-based query planning. - Benchmarked relevance, grounding, and latency across retriever and model settings, then tested the service at 25 concurrent users with sub-500 ms response time. Added OpenTelemetry tracing to make retrieval and service failures easier to diagnose.	

Teaching Experience

Algorithms (CS 310) , Teaching Assistant, LUMS	Aug. – Dec. 2023
Introduction to Blockchain (CS 3812) , Teaching Assistant, LUMS	Jan. – May 2023
Topics in Blockchain Technology (CS 589) , Teaching Assistant, LUMS	Aug. – Dec. 2022
Computational Problem Solving (CS 100) , Teaching Assistant, LUMS	Aug. – Dec. 2022

Academic Service & Mentoring

Reviewer , SemEval 2026 Tasks 6 & 8; JUDGE 2026 (NeurIPS Workshop)	2026
Presenter , ACL 2026 (virtual), SemEval 2026 Task 6 and Task 8 papers	2026
Mentor , GradApp Lab; LUMS Alumni Mentorship Program	July 2024 – Present

Technical Skills

Research methods: human annotation and evaluator calibration, equivalence testing (TOST), bootstrap confidence intervals, mixed-effects modeling, reproducibility auditing, preregistered/seed-controlled experimentation	
ML/NLP: PyTorch, Hugging Face Transformers, PEFT (LoRA/QLoRA), vLLM, DeepSpeed, scikit-learn, LangGraph; OpenSearch, FAISS, BM25/SPLADE, dense/hybrid retrieval, cross-encoder reranking	
Systems: Python, TypeScript, SQL, C/C++, Bash; SLURM/HPC, Docker, Git, PostgreSQL, Linux; FHIR R4, CQL/ELM, SMART Backend Services, MCP	